TASK 2

Interactive Power BI Executive Sales & Profit Dashboard

Solved analysis report and Power BI implementation guide

$2,297,200.86	$286,397.02	12.47%	5,009
TOTAL SALES	TOTAL PROFIT	PROFIT MARGIN	DISTINCT ORDERS

Dataset: Superstore sales CSV **Valid transaction rows:** 9,994 **Columns:** 21

Coverage: 03 Jan 2015 to 30 Dec 2018 **Customers:** 793 **Quantity:** 37,873

What this PDF includes

Dataset cleaning summary, executive KPIs, sales and profit findings, regional/category analysis, recommended dashboard layout, data model, DAX measures, and a submission checklist.

Important: This is the completed analysis/report PDF. The editable Power BI **.pbix** file must still be created and saved in Power BI Desktop; a PDF cannot replace the required PBIX deliverable.

1. Executive Summary

The Superstore dataset contains 9,994 valid order-line records after removing rows with missing Order Date. Across the period 2015–2018, the business generated strong overall sales and positive profit. Technology generated the highest category sales and profit; the West region generated the highest sales.

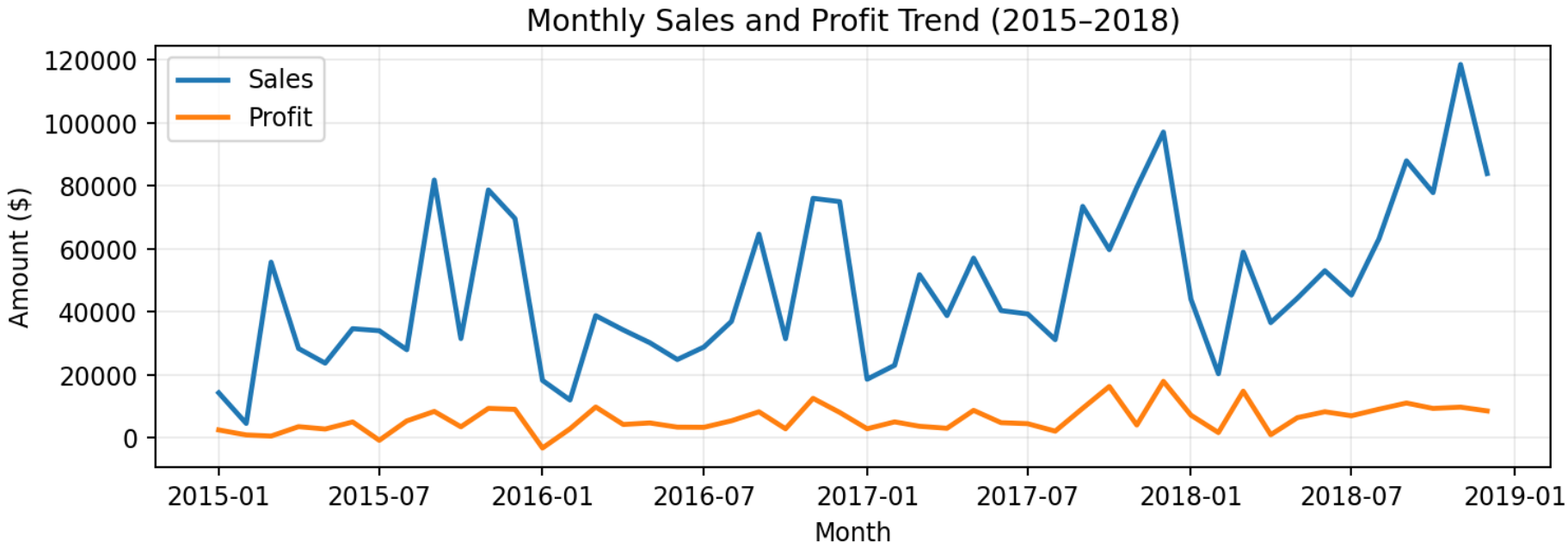
Metric	Result	Meaning
Total Sales	$2,297,200.86	Total revenue across valid records
Total Profit	$286,397.02	Net profit recorded in the dataset
Profit Margin	12.47%	Total Profit ÷ Total Sales
Distinct Orders	5,009	Unique Order IDs
Distinct Customers	793	Unique Customer IDs
Total Quantity	37,873	Units sold
Average Sales per Order	$458.61	Total Sales ÷ Distinct Orders

Data quality notes

The source file has 10,800 rows, but 806 rows have no usable Order Date and were excluded. The cleaned analysis therefore uses 9,994 rows. There are 11 missing Postal Code values among the valid records; these should remain blank or be labelled Unknown rather than deleting otherwise valid sales.

Recommended cleaning: parse Order Date and Ship Date as Date; set Sales, Profit and Discount as Decimal Number; set Quantity as Whole Number; set Postal Code as Text; remove rows where Order Date is blank; remove duplicates only if the business confirms they are duplicate transactions. Do not remove negative Profit values: they represent losses.

2. Sales and Profit Trend

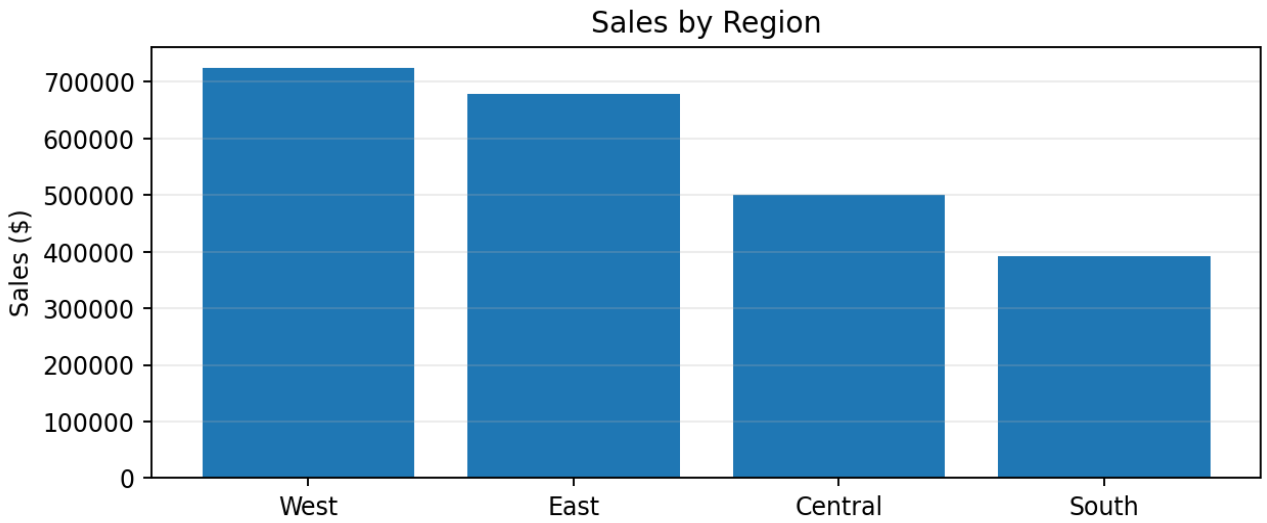


Yearly performance

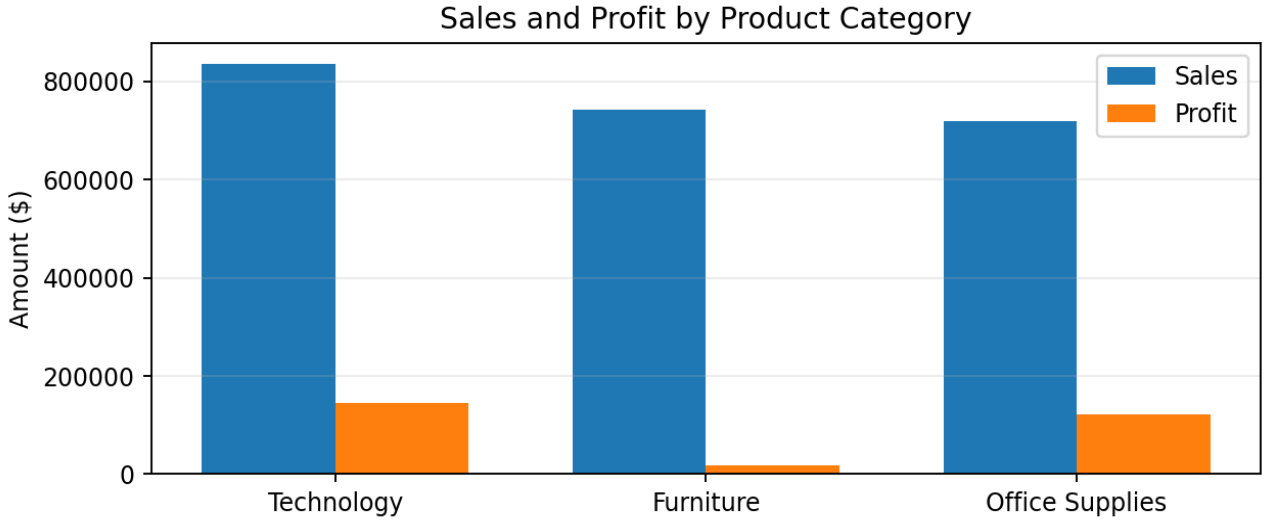
Year	Sales	Profit	Sales growth vs previous year
2015	$484,247.50	$49,543.97	—
2016	$470,532.51	$61,618.60	-2.83%
2017	$609,205.60	$81,795.17	29.47%
2018	$733,215.26	$93,439.27	20.36%

Insight: Sales increased from 2016 onward and reached the highest annual level in 2018. Use a continuous date axis in Power BI so monthly trends are displayed chronologically.

3. Regional and Category Performance

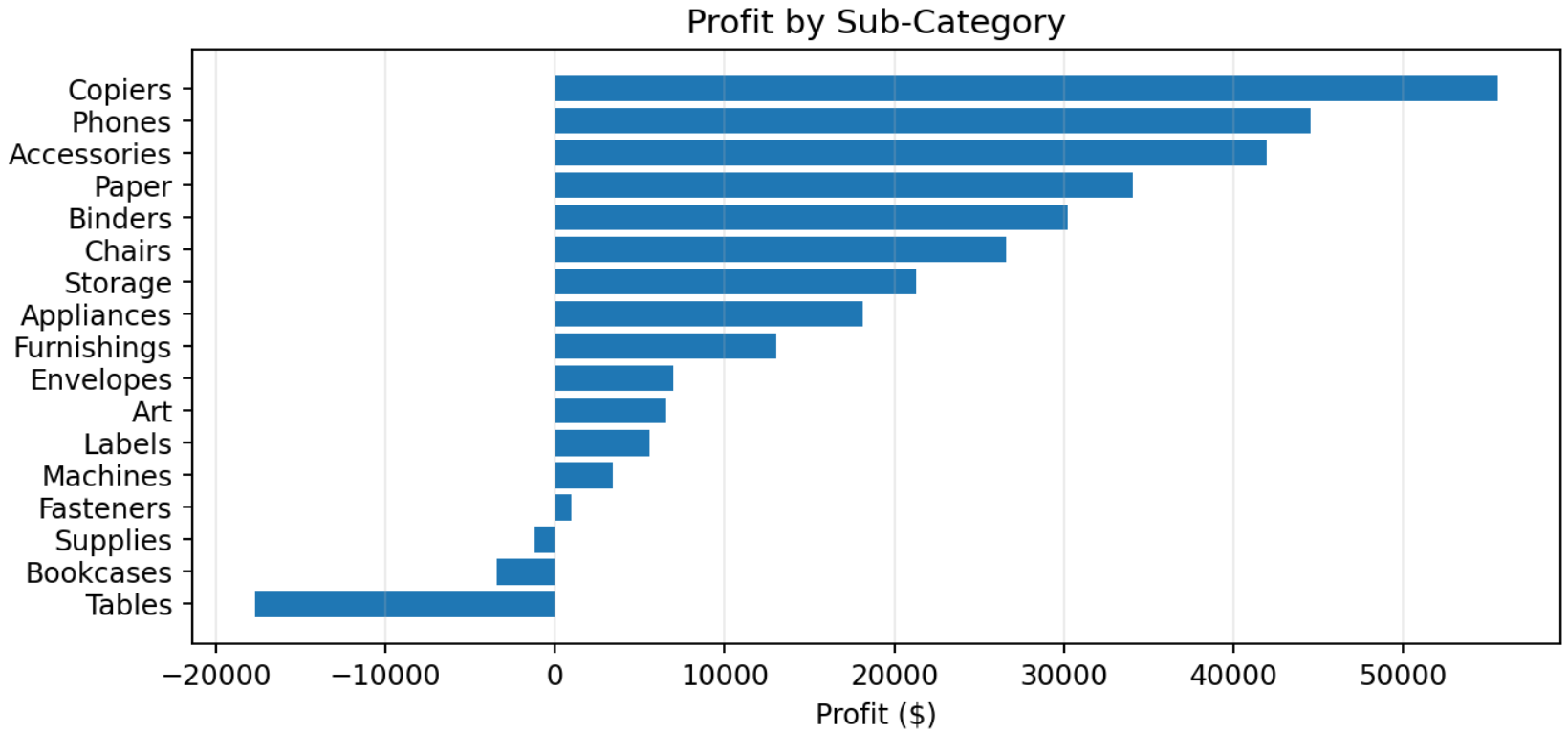


Region	Sales	Profit	Profit Margin
West	$725,457.82	$108,418.45	14.94%
East	$678,781.24	$91,522.78	13.48%
Central	$501,239.89	$39,706.36	7.92%
South	$391,721.91	$46,749.43	11.93%



Category	Sales	Profit	Profit Margin
Technology	$836,154.03	$145,454.95	17.40%
Furniture	$741,999.80	$18,451.27	2.49%
Office Supplies	$719,047.03	$122,490.80	17.04%

4. Product Profitability



Top sub-categories by sales

Sub-Category	Sales	Profit
Phones	$330,007.05	$44,515.73
Chairs	$328,449.10	$26,590.17
Storage	$223,843.61	$21,278.83
Tables	$206,965.53	$-17,725.48
Binders	$203,412.73	$30,221.76

Sub-Category	Sales	Profit
Machines	$189,238.63	$3,384.76
Accessories	$167,380.32	$41,936.64
Copiers	$149,528.03	$55,617.82

Attention area: the lowest-profit sub-categories should be reviewed for discounting, returns, product cost, and pricing. A high sales value does not always mean high profitability.

5. Power BI Data Model and Dashboard Plan

Recommended star schema

FACT ORDERS (one row per order line): Row ID, Order ID, Order Date, Ship Date, Customer ID, Product ID, Sales, Profit, Quantity, Discount, Ship Mode, Region, State, City and Postal Code.
DIM CUSTOMER: Customer ID (unique), Customer Name, Segment.
DIM PRODUCT: Product ID (unique), Product Name, Category, Sub-Category.
DIM DATE: Date (unique), Year, Quarter, Month, Month Number.

Dimension table	Relationship to Fact Orders	Cardinality
Dim Customer	Dim Customer[Customer ID] → Fact Orders[Customer ID]	1 to many
Dim Product	Dim Product[Product ID] → Fact Orders[Product ID]	1 to many
Dim Date	Dim Date[Date] → Fact Orders[Order Date]	1 to many

Create the dimension tables in Power Query using Reference, retain only the listed columns, and remove duplicates from each dimension key. Create the Date table in DAX or Power Query. Use single-direction filtering from each dimension to the fact table. Mark Dim Date as the date table.

Dashboard page layout

Dashboard area	Recommended visual	Fields
Top KPI row	4 Card visuals	Total Sales, Total Profit, Profit Margin %, Total Orders
Top/side filters	3 Slicers	Dim Date[Date], Region, Category
Middle left	Line chart	Axis: Month; Values: Total Sales; Legend: Year
Middle right	Map or bar chart	Location: State; Size/Values: Total Sales; Tooltips: Profit
Bottom left	Treemap	Group: Category; Details: Sub-Category; Values: Total Sales
Bottom right	Waterfall / bar chart	Category: Month or Sub-Category; Values: Total Profit

6. DAX Measures (Copy into Power BI)

First rename your cleaned transaction table to **Fact Orders** in Power Query, or replace the table name below with the exact name shown in your Fields pane.

Total Sales

```
Total Sales = SUM('Fact Orders'[Sales])
```

Total Profit

```
Total Profit = SUM('Fact Orders'[Profit])
```

Profit Margin %

```
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
```

Total Orders

```
Total Orders = DISTINCTCOUNT('Fact Orders'[Order ID])
```

Total Quantity

```
Total Quantity = SUM('Fact Orders'[Quantity])
```

Average Order Sales

```
Average Order Sales = DIVIDE([Total Sales], [Total Orders], 0)
```

Previous Year Sales

```
Previous Year Sales = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Dim Date'[Date]))
```

YoY Sales Growth %

```
YoY Sales Growth % = DIVIDE([Total Sales] - [Previous Year Sales], [Previous Year Sales], 0)
```

Format Profit Margin % and YoY Sales Growth % as Percentage. Format Total Sales, Total Profit and Average Order Sales as Currency. The YoY measure requires an active relationship from Dim Date[Date] to Fact Orders[Order Date].

7. Business Insights and Recommendations

1. Overall performance: total sales are $2,297,200.86 with total profit of $286,397.02 and a 12.47% profit margin.

2. Regional leader: West is the highest-sales region, contributing 31.58% of total sales.

3. Category leader: Technology is the highest-sales category and also has the highest total category profit.

4. Time trend: 2018 has the highest yearly sales ($733,215.26).

5. Profitability: Phones is the lowest-profit sub-category by total profit ($44,515.73); review pricing, discount and cost drivers.

Recommended actions

• Monitor profit margin alongside sales; do not evaluate performance using revenue alone.
• Investigate high-discount orders and loss-making products/sub-categories.
• Compare regional results using both sales and profit.
• Use date and region slicers to support executive drill-down and decision-making.

8. Final Submission Checklist

Item	Expected output	Status
Cleaned dataset	9,994 valid rows; blank Order Date rows excluded	Analysis completed
Data model	Fact Orders + Dim Customer + Dim Product + Dim Date	Build in Power BI
DAX measures	Sales, Profit, Margin, Orders, Quantity, YoY Growth	Formulas provided
Interactive report	Cards, date/region/category slicers, charts and treemap	Build in Power BI
Power BI project	Task_2_Executive_Sales_Profit_Dashboard.pbix	Save from Power BI Desktop
PDF report	Task_2_Executive_Sales_Profit_Dashboard_Solved_Report.pdf	This document

Submission note: Submit both the editable PBIX file and this PDF if the internship portal requests both. This report provides the completed dataset analysis and implementation instructions, but it does not contain an editable PBIX file.